1. Backpropagation

input = $x = [x_1, x_2, x_3] = [0.5, 0.5, 0.1]$

ground truth = $y = [y_1, y_2] = [0.9, 0.1]$

activation sigmoid = $1 / 1+e^{-z}$

1.1 Forward pass

$z_1^1 = x_1 w_{1,1}^1 + x_2 w_{1,2}^1 + x_3 w_{1,3}^1 + b_1^1$

$= 0.5 \cdot 0.1 + 0.5 \cdot 0.2 + 0.1 \cdot 0.5 + 0.2$

$= 0.05 + 0.1 + 0.05 + 0.2 = 0.4$

$\Rightarrow h_1 = \sigma(0.4) = \frac{1}{1+e^{-0.4}} \simeq 0.5987$

$z_2^1 = x_1 w_{2,1}^1 + x_2 \cdot w_{2,2}^1 + x_3 w_{2,3}^1 + b_2^1$

$= 0.5 \cdot 0.4 + 0.5 \cdot 0.1 + 0.1 \cdot 0.5 + 0.2$

$= 0.2 + 0.05 + 0.05 + 0.2 = 0.5$

$\Rightarrow h_2 = \sigma(0.5) = \frac{1}{1+e^{-0.5}} \simeq 0.6225$

1.2 Output layer

$z_1^2 = h_1 w_{1,1}^2 + h_2 w_{1,2}^2 + b_1^2$

$= 0.6 \cdot 0.5987 + 0.1 \cdot 0.6225 + 0.2$

$= 0.3592 + 0.0622 + 0.2 = 0.6215$

$y_1 = \sigma(0.6215) \simeq 0.6506$

$z_2^2 = h_1 \cdot w_{2,1}^2 + h_2 w_{2,2}^2 + b_2^2$

$= 0.5987 \cdot 0.5 + 0.6225 \cdot 0.2 + 0.1$

$= 0.2994 + 0.1245 + 0.1 = 0.5238$

$y_2 = \sigma(0.5238) \simeq 0.6280$

2 Loss

$L = \frac{1}{2} \sum_{i=1}^{2} (y_i - \hat{y}_i)^2 = \frac{1}{2} [(0.6506 - 0.9)^2 + (0.6280 - 0.1)^2]$

$= \frac{1}{2} [(-0.2494)^2 + (0.528[redacted])^2]$

$= \frac{1}{2} (0.0622 + 0.2788) \simeq 0.1705$

2. Backword pass

$$\frac{\partial L}{\partial w^1_{1,3}} = \overset{①}{\frac{\partial L}{\partial h_1}} \; \overset{②}{\frac{\partial h_1}{\partial z_1^1}} \; \overset{③}{\frac{\partial z_1^1}{\partial w^1_{1,3}}} = 0.02762 \cdot 0.2403 \cdot 0.1 = 0.0006637$$

① $$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial z_1^2} \cdot w^2_{1,1} + \frac{\partial L}{\partial z_2^2} \cdot w^2_{2,1} = -0.0567 \cdot 0.6 + 0.1233 \cdot 0.5 =$$

$$-0.03402 + 0.06165 = \boxed{0.02762}$$

$$\frac{\partial L}{\partial z_1^2} = \frac{\partial L}{\partial y_1} \frac{\partial y_1}{\partial z_1^{(1)}} = \frac{\partial}{\partial y_1}\left(\frac{1}{2}(y_1 - \hat{y}_1)^2\right) \underbrace{\frac{\partial y_1}{\partial z_1^2}}_{\sigma(z_1^1)(1-\sigma(z_1^1))} = (y_1 - \hat{y}_1) \cdot y_1(1-y_1) =$$

$$= (0.6506 - 0.9)(0.6506) \cdot (1 - 0.6506) = -0.2494 \cdot 0.6506 \cdot 0.3494$$

$$= \underline{-0.0567}$$

$$\frac{\partial L}{\partial z_2^2} = \frac{\partial L}{\partial y_2} \frac{\partial y_2}{\partial z_2^2} = \frac{\partial}{\partial y_2}\left(\frac{1}{2}(y_2 - \hat{y}_2)^2\right) \frac{\partial y_2}{\partial z_2^2} = (y_2 - \hat{y}_2) \cdot (y_2)(1 - y_2) =$$

$$= (0.6280 - 0.1) \cdot (0.6280)(1 - 0.6280)$$

$$= (0.5280) \cdot (0.6280) \cdot (0.3720)$$

$$= \underline{0.1233}$$

~~2/4~~ ② $$\frac{\partial h_1}{\partial z_1^1} = h_1(1 - h_1) = 0.5987(1 - 0.5987) = 0.5987 \cdot 0.4013 = \boxed{0.2403}$$

③ $$\frac{\partial z_1^1}{\partial w^1_{1,3}} = \frac{\partial (x_1 w^1_{1,1} + x_2 w^1_{1,2} + x_3 w^1_{1,3} + b_1^1)}{\partial w^1_{1,3}} = x_3 = \boxed{0.1}$$

4. weight update

$$w^1_{1,3} = w^1_{1,3} - \eta \cdot \frac{\partial L}{\partial w^1_{1,3}} = 0.5 - 0.1 \cdot (0.0006637) = 0.5 - 0.0000663 7$$

$$= \underline{0.49993363}$$